

CE902 Professional Practice and Research Methodology (2020)

Worked solutions for all questions visible in the uploaded paper

Important source note: the uploaded exam states that Section A is unavailable. This document therefore solves the whole visible paper, i.e. Section B Questions 18–22.

Method and presentation

Each answer below shows the reasoning, formulas and intermediate steps used to reach the result. For calculation questions, values are rounded sensibly to two decimal places unless an exact expression is clearer. For discursive questions, the answer is structured as an examinable response rather than a bare bullet list.

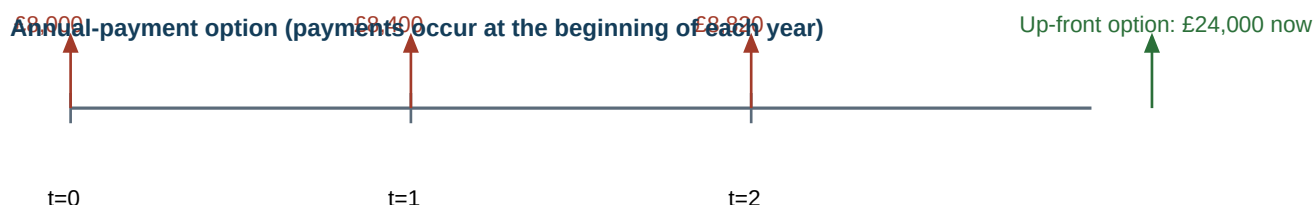
Question 18 – Lease decision using Net Present Value

The decision must be based on present value rather than nominal cash paid, because the two payment schemes occur at different times. The question provides the standard present value formula:

$$PV = X / (1 + r)^t$$

where X is the payment being evaluated, r is the discount rate (bank interest rate), and t is the number of years from today. Because the annual lease payments are made at the **beginning** of each year and are increased by inflation, the three nominal payments are:

Year 0: £8,000; Year 1: £8,000 × 1.05 = £8,400; Year 2: £8,400 × 1.05 = £8,820.



Option	Present-value expression	NPV cost
Pay up-front now	£24,000 (already at time 0)	£24,000.00
Annual payments, bank rate 3%	£8,000 + £8,400/1.03 + £8,820/1.03 ²	£24,469.04
Annual payments, bank rate 9%	£8,000 + £8,400/1.09 + £8,820/1.09 ²	£23,130.04

Interpretation at 3% bank interest

The up-front option has an NPV cost of £24,000.00. The annual-payment option has an NPV cost of £24,469.04. Since a cost-minimisation problem should select the **lower** present-value cost, paying up-front is better by £469.04. The intuition is that inflation at 5% is higher than the discount rate at 3%, so the later payments grow faster than they are discounted.

Interpretation at 9% bank interest

If the bank rate rises to 9%, the annual-payment option falls to an NPV cost of £23,130.04. This is lower than the up-front cost of £24,000.00 by £869.96. Therefore the decision **does change**: with the higher discount rate, deferring payment becomes financially preferable because future payments are discounted much more heavily.

Question 19 – Referencing and citation

Part (a): deciding whether a citation is needed and where it should be placed

Extracts	Is a citation needed?	Where it should go
I	Yes	After each study claim. The first sentence reports one study and the second reports a later study, so e
II	Yes	At the end of the sentence containing the climate statistics, because the date rankings and the attribut
III	Yes	At the end of the sentence (or immediately after the cited proposition), because the wording 'It has be

Part (b): three reasons for referencing other people's work

- To acknowledge intellectual ownership and avoid plagiarism.
- To allow the reader to locate, verify and evaluate the original source for themselves.
- To strengthen the credibility of the assignment by showing that claims are based on evidence rather than unsupported opinion.

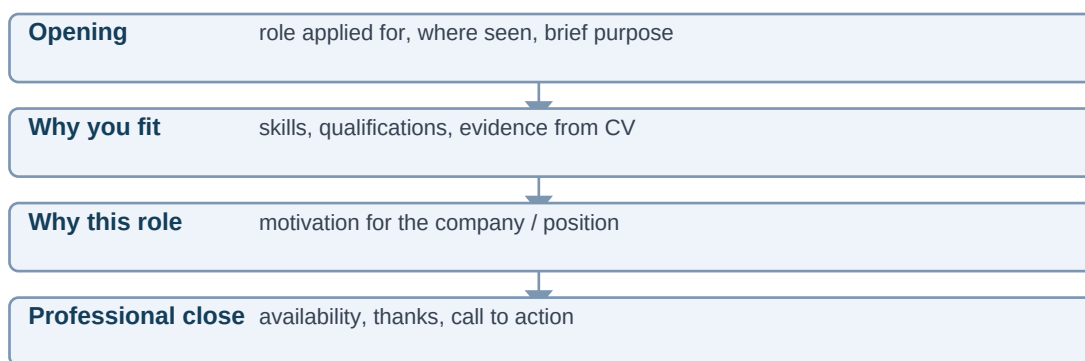
Part (c): two examples of material that normally does not need referencing

- Common knowledge, for example facts that are widely known and undisputed.
- Your own original observations, experience, reflections or data that you generated yourself (provided they are genuinely your own and not derived from a source).

A useful check is this: if a reader could reasonably ask "where did that come from?", then a citation is normally required.

Question 20 – Job application documents and recruitment criteria

Part (a): competent covering letter structure



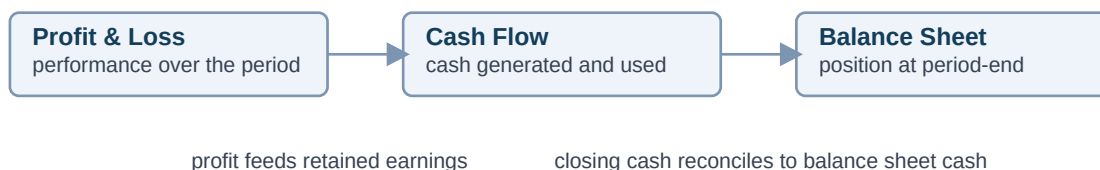
A competent covering letter should open professionally, naming the role being applied for and where the vacancy was seen. It should then connect the applicant's skills, qualifications and relevant experience to the requirements of the role, using a few concise examples rather than repeating the CV verbatim. A strong middle section explains why the applicant wants *this* role and *this* employer, showing motivation and fit. The letter should close politely by confirming interest, availability for interview, and appreciation for the employer's time. The overall tone should be clear, targeted, concise and businesslike.

Part (b): four qualities likely to be sought in the Graduate Software Developer role

Criterion	Why it follows from the advert
Technical software-development potential	The advert emphasises developing software applications and integrated systems, so evidence of programming ability and sound software fundamentals would matter.

Ability to learn quickly and work with current technology	The role promises exposure to 'some of the latest technology in the market today', so curiosity and willingness to upskill are important.
Teamwork and communication	The candidate will work alongside experienced developers and systems integrators in a fast-paced environment, so collaboration skills are essential.
Professionalism and adaptability	The role supports large manufacturing clients, so reliability, organisation and the ability to work in an industrial/business setting would be attractive.

Question 21 – Purpose of the major annual-report statements



Statement	Main purpose	Typical question answered
Balance Sheet	Shows assets, liabilities and equity at a specific date	What does the company own and owe right now?
Profit & Loss Account	Shows revenue, costs and profit/loss over the accounting period	Did the company trade profitably during the period?
Cash Flow Statement	Shows actual cash inflows and outflows from operating, investing and financing activities	How did the company fund and finance its business?

These reports are connected rather than independent. Profit from the Profit & Loss Account contributes to retained earnings within equity on the Balance Sheet, while the closing cash balance in the Cash Flow Statement must reconcile with the cash figure shown on the Balance Sheet. Used together, the three statements let an analyst judge profitability (P&L); liquidity (Cash Flow) and financial position/solvency (Balance Sheet). A company can be profitable but cash-poor, or cash-rich but weakly profitable, so a single statement is not enough for a sound judgement.

Question 22 – Essay conclusion order and report introduction

Part (a): ordering the conclusion sentences

A strong conclusion usually restates the central argument, briefly summarises the key support, acknowledges any important limitation, and then ends with a final evaluative or forward-looking sentence. Using that logic, the most coherent sequence is:

D → B → A → C

Sentence D restates the essay's overall judgement. Sentence B summarises the main supporting finding. Sentence A introduces the limitation or caveat. Sentence C gives the final balanced recommendation for future work, which is an appropriate final sentence.

Part (b): purpose of the Introduction section to a report

The purpose of an introduction is to orient the reader: it establishes what the report is about, why it has been written, and how the reader should approach what follows. Two things that should normally appear in an introduction are:

- brief background/context for the topic and the purpose or aim of the report;
- the scope of the report, often with a short statement of how the report is organised.

Final note: this worked solution is limited to the material present in the uploaded source. Section A was marked as unavailable in the exam PDF, so no Section A solutions could be derived from the file.